

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

Course: B. Tech.

Branch : Information Technology

Semester : IV

Subject Code & Name:

BTITC403* & Discrete Mathematics

Max Marks: 60

Date: 24/12/2024

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Which of the following is a tautology? a. $P \vee Q$ b. $P \vee \neg P$ c. $P \wedge Q$ d. $P \wedge \neg P$	CO1	1
2	Let $A = \{1,2,3\}$ and $B = \{4,5\}$. How many relations can be defined from A to B? a. 6 b. 8 c. 32 d. 64	CO1	1
3	A simple undirected graph has 7 vertices and 10 edges. The sum of the degrees of all vertices is: a. 7 b. 10 c. 14 d. 20	CO3	1
4	If $f: A \rightarrow B$ is one-to-one, then which of the following is true? a. $ A \leq B $ b. $ A > B $ c. $ A = B $ d. $ A \geq B $	CO1	1
5	A die is rolled. What is the probability of getting a number greater than 4? a. $1/2$ b. $1/3$ c. $2/3$ d. $1/6$	CO2	1
6	How many 3-digit numbers can be formed using the digits 1,2,3,4 if repetition is not allowed? a. 24 b. 64 c. 120 d. 720	CO2	1
7	The recurrence relation $T(n) = 2T(n/2) + n$ has a solution of: a. $O(\log n)$ b. $O(n)$ c. $O(n \log n)$ d. $O(n^2)$	CO2	1
8	The dual of the Boolean expression $A + 0 = A$ is: a. $A \cdot 1 = A$ b. $A \cdot 0 = 0$ c. $A + 1 = 1$ d. $A + A = A$	CO2	1
9	Which of the following is a group under addition? a. Natural Numbers b. Integers c. Positive Rational Numbers d. Non-zero Real Numbers	CO4	1
10	The minimum number of colours required to colour a complete graph K_4 is: a. 2 b. 3 c. 4 d. 5	CO3	1

11	How many ways are there to distribute 5 indistinguishable balls into 3 distinguishable boxes, such that no box is empty?				CO2	1
	a. 5	b. 6	c. 7	d. 10		
12	Which of the following pairs of graphs are isomorphic?					1
	a. Two cycles with 4 vertices each	b. A cycle with 4 vertices and a complete graph K_4	c. Two complete graphs K_3	d. A cycle with 5 vertices and a complete graph K_5		
Q. 2	Solve the following.					12
A)	Let A, B and C be sets such that: $A =50, B =30, C =20$ $A \cap B =10, A \cap C =5, B \cap C =8$ $A \cap B \cap C =3$ Using the Principle of Inclusion-Exclusion, calculate: $A \cup B \cup C$ - The number of elements belonging only to set A.				CO1	6
B)	Translate the following English statement into a propositional logic expression. Then, prove its validity using a truth table or rules of inference. Statement: https://www.batuonline.com "If a student studies hard, then they pass the exam. If they pass the exam, then they get a certificate. Therefore, if a student studies hard, they get a certificate."				CO1	6
Q.3	Solve the following.					12
A)	Prove by mathematical induction that for all $n \geq 1$: $1^3+2^3+3^3+\dots+n^3=[n(n+1)/2]^2$				CO2	6
B)	A recursive function $f(n)$ is defined as follows: $f(n) = \begin{cases} 1 & \text{if } n=0, \\ 2 \cdot f(n-1) + 1 & \text{if } n>0. \end{cases}$ - Write the first four values of $f(n)$ (for $n=0, 1, 2, \dots$). - Derive a closed-form expression for $f(n)$ and verify using induction.				CO2	6
Q. 4	Solve Any Two of the following.					12
A)	Prove that in any group of 13 people, at least 2 of them were born in the same month.				CO3	6
B)	Solve the recurrence relation $a_n=3a_{n-1}-2a_{n-2}$ with initial conditions $a_0=2, a_1=5$.				CO3	6

C)	How many distinct 6-letter words can be formed using the letters of the word BANANA?	CO3	6
Q.5	Solve Any Two of the following.		12
A)	Consider the directed graph represented by the adjacency matrix below: $A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$ - Find the transitive closure of the relation using Warshall's Algorithm. - Write the final adjacency matrix representing the transitive closure.	CO1	6
B)	Prove that the set of integers Z under addition forms a group, but the set of positive integers Z^+ under addition does not.	CO4	6
C)	Using Fermat's Little Theorem, compute $7^{100} \text{ mod } 13$.	CO4	6
Q. 6	Solve Any Two of the following.		12
A)	Consider the weighted graph below: Vertices: A, B, C, D Edges: (A, B, 4), (A, C, 2), (B, C, 1), (B, D, 5), (C, D, 8) - Find the shortest path from vertex A to vertex D using Dijkstra's Algorithm. - Write the sequence of vertices in the shortest path and its total weight.	CO3	6
B)	Prove whether the following graph is planar or not: K_5 , the complete graph on 5 vertices.	CO3	6
C)	Find the Minimal Spanning Tree (MST) for the following graph using Prim's Algorithm: Vertices: A, B, C, D, E Edges with weights: (A, B, 1), (A, C, 5), (B, C, 2), (B, D, 4), (C, D, 6), (D, E, 3)	CO3	6
*** End ***			